AccessMatrix Performance Benchmark Report - 6.0.1-GA (AI)

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1. Introduction

This document contains the performance result of AccessMatrix Server performing certain use cases as stated in the sections. The performance number is an indication of what AccessMatrix Server can perform based on the test configuration. The goal of the stress test is to be repeatable; thus, normally default configuration (JVM heap size, connection pool, event worker pool) is used unless stated otherwise. However, there could be cases where deployment in certain environment may produce different result. Thus, the information in this document can be used for planning and sizing estimation. It is still recommended to do actual stress test based on the project's use cases.

History of changes

- **September 2025:** Added "E2EEA Thales PSE3 (PTK C) - AuthenticationService - 1FA - E2EE Password(PQC)"
- **February 2025:** Added "OpenShift/Kubernetes Scalability" under "UAS"
- **February 2024:** Added "VASCO Multi-Device - TokenService - verifyDigitalSignature"
- **February 2024:** Added "VASCO Multi-Device - TokenService - verifyMessageSignature (REST)"
- **February 2024:** Added "VASCO Multi-Device - TokenService - Activation(Redis Cluster)"
- **January 2024:** Added "Session - Verify"
- **January 2024:** Added "Session - Invalidate"
- **December 2023:** Added "FIDO2 - Token Service - assertionFinish"
- **April 2023:** Added "SMSOTP Token - TokenService - Verify OTP (REST)"
- **August 2022:** Added "OIDC - Introspection Endpoint - OAuthProxy and AM6 as OpenID Provider"
- **April 2022:** Added "E2EEA Utimaco CryptoServer - AuthenticationService - 1FA - E2EE Password"
- **March 2022:** Added "VASCO Multi-Device - TokenService - verify OTP (REST)"
- **March 2022:** Added "E2EEA Thales PSE3 (PTK C) - AuthenticationService - 1FA - E2EE Password"

2. Benchmark Result

Default testing configuration (unless stated otherwise):

- **Web App Server:** Bundled Apache Tomcat on JRE 11 (server 64-bit)
- **Java Heap Size:** 1 GB (default of bundled Apache Tomcat)
- **TLS:** All API calls are tested using TLS connection (AES-256 cipher suite) with client authentication
- **Tomcat Connector:** NIO2 connector
- **AM6 API:** REST API or XML-RPC API Java Wrapper
- **User Store:** Default store
- **CPU:** CPU utilization is targeted to use 75% or less on AM server (unless stated otherwise)

Benchmark results:

- Core
- UAS
- UAM
- USO

2.1. Core

Tomcat Connectors

Hardware

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	4 GB
DB - Oracle 12c Standard 2	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	6 GB

Test Result

1FA password authentication w/ audit w/ history
Password hash algo: SHA-256

No significant difference in term of throughput, memory usage, the response time when comparing BIO and NIO. Basically, NIO, as designed, does not tie one thread per connection. It has a pool of connections and a separate pool of worker threads. Thus, it allows for serving more connections. The idle connection does not occupy the worker thread. Whereas for BIO, one thread is tied to one connection. Thus, the number of maximum connections is directly tied to the number of worker threads. In BIO, idle connection occupies the worker thread.

Whether it is significant to use NIO or BIO depends on the maximum number of connections from the AM6 clients, and also whether the load is likely to be distributed or not. In general, NIO is more efficient. In Apache Tomcat 7, BIO is the default connector. In Apache Tomcat 8, NIO is the default connector.

General Purpose Encrypt and Decrypt API (Luna SA7 HSM)

Hardware

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	8GB

Database and IO is not significant.

Configuration

Key in HSM	AES 256-bit
Data length	32 bytes clear text
Data length	512 bytes clear text

Test Result

Calls km/encrypt and km/decrypt.

AM CPU usage: 50%

Throughput/sec	32 byte data	512 byte data
/km/encrypt	1250	1200
/km/decrypt	1250	1200

General Purpose Encrypt and Decrypt API (non-HSM System Key Provider)

Hardware

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	8GB

Database and IO is not significant.

Configuration

Key	AES 256-bit
Data length	32 bytes clear text
Data length	512 bytes clear text

Test Result

Calls km/encrypt and km/decrypt.

AM CPU usage: 60%

Throughput/sec	32 byte data	512 byte data
/km/encrypt	1800	1700
/km/decrypt	1800	1700

2.2. UAS

Password - AuthenticationService - 1FA - Password

Hardware

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	4 GB
DB - Oracle 12c Standard 2	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	6 GB

Configuration

Password hash algo	SHA-256
Number of realm accounts	2
Number of login accounts	2

Test Result

Session Persistence: Off

	w/ audit w/ history		w/o audit w/o history	
Num of users in DB	500K	1000K	500K	1000K
Throughput	420 authn/sec	420 authn/sec	750 authn/sec	750 authn/sec

Session Persistence: On

	w/ audit w/ history		w/o audit w/o history	
Persistence Option	DB	Redis	DB	Redis
Throughput	314.4authn/sec	286.04 authn/sec	559.3 authn/sec	566.7 authn/sec

E2EEA Gemalto LunaEFT2 (PTKComm) - AuthenticationService - 1FA - E2EE Password

Hardware

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	4 GB
DB - Oracle 12c Standard 2	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	6 GB
HSM	
HSM	Safenet LunaEFT2 PL280
HSM	Safenet LunaEFT2 PL1200

Configuration

Password hash algo	SHA-256
RSA-OAEP hash algo for RSA-1024	SHA1
RSA-OAEP hash algo for RSA-2048	SHA-256
Storage key (KTPV)	3DES
Server random number length	16 bytes

Number of realm accounts	1
Number of login accounts	1
Number of users in DB	1200K

Test Scenario

- Preauthentication to generate E2EE session and to get a random number and public key.
- Login to verify user credential.

Test Result

RSA Key Size	**1024**	**2048**
Throughput (PL280)	85 authn/sec	55 authn/sec
Throughput (PL1200)	125 authn/sec	65 authn/sec
Preauth avg response time (PL280)	255 ms	-
Preauth avg response time (PL1200)	80 ms	210 ms
Login avg response time (PL280)	180 ms	-
Login avg response time (PL1200)	75 ms	180 ms

Notes:

- Bottleneck is at the HSM. AM6 and DB CPU usages are considered low, 30% or less.
- Higher HSM PL level does not scale in a linear manner for E2EEA (e.g. EFT2 1200 is not 4 times faster than PL280).
- RSA-OAEP hash algo has negligible performance impact (SHA1 vs SHA-256 vs SHA-512).
- AM server tested for pure verification (RSA-2048) on PL1200 HSM reaches a throughput of 83 authentications/sec.
- Standalone native utility calling PL1200 HSM via Safenet PTKComm native API for pure verification (RSA-2048, EE3005) reaches a throughput of 83 authentications/sec.
- PTKComm load balance is limited to the throughput of one HSM, although multiple HSMs are used for load balancing.

E2EEA Gemalto LunaEFT2 (Socket API, Firmware 2.3.0) - AuthenticationService - 1FA - E2EE Password

Note: The E2EEA Gemalto LunaEFT2 via socket API is only available from AccessMatrix version 5.6.0.

Hardware

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	4 GB
DB - Oracle 12c Standard 2	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	6 GB
HSM	
HSM	Safenet LunaEFT2 PL60 x1
HSM	Safenet LunaEFT2 PL280 x2
HSM	Safenet LunaEFT2 PL1200 x2
Load Balancer	
Load Balancer	HAproxy hosted in the same machine as AM6

Configuration

Password hash algo	SHA-256
RSA-OAEP hash algo for RSA-1024	SHA1
RSA-OAEP hash algo for RSA-2048	SHA-256
Storage key (KTPV)	3DES

Server random number length	16 bytes
AM connection pool size (1 HSM)	2
AM connection pool size (2 HSMs via load balancer)	4

Number of realm accounts	1
Number of login accounts	1
Number of users in DB	1200K

Test Scenario

- Preauthentication to generate E2EE session and to get a random number and public key.
- Login to verify user credential.

Test Result for Multiple HSMs

Stress is done with login module setting not to cache public key in AM and to get server random from HSM.

	1 HSM		2 HSMs	
RSA Key Size	**1024**	**2048**	**1024**	**2048**
Throughput (PL60)	19 authn/sec	19 authn/sec	(not tested)	(not tested)
Throughput (PL280)	85 authn/sec	55 authn/sec	160 authn/sec	110 authn/sec
Throughput (PL1200)	150 authn/sec	70 authn/sec	240 authn/sec	130 authn/sec
Preauth avg response time (PL60)	170 ms	160 ms	(not tested)	(not tested)
Preauth avg response time (PL280)	120 ms	100 ms	50 ms	90 ms
Preauth avg response time (PL1200)	60 ms	100 ms	40 ms	70 ms
Login avg response time (PL60)	90 ms	105 ms	(not tested)	(not tested)
Login avg response time (PL280)	90 ms	120 ms	50 ms	90 ms
Login avg response time (PL1200)	70 ms	110 ms	90 ms	80 ms

Notes:

- Bottleneck is at the HSM. AM6 and DB CPU usages are considered low, 50% or less for 2048-bit. For 1024-bit, DB CPU usage is high.
- Higher HSM PL level does not scale in a linear manner for E2EEA (e.g. EFT2 1200 is not 4 times faster than PL280).
- RSA-OAEP hash algo has negligible performance impact (SHA1 vs SHA-256 vs SHA-512).
- It is recommended to set connection pool size at AM (AM ==> load balancer) to 2 connections per HSM.
- HSM software version 2.3.0-368 has a slight performance improvement of 5%.

Test Result for public key caching and Java SecureRandom

Stress test is done on 1 HSM only and 2048-bit RSA key.

Public key	**not cached**	**cached**	
Random generation	**HSM**	**HSM**	**AM**
Throughput including preauth (PL60)	19 authn/sec	28 authn/sec	55 authn/sec
Throughput including preauth (PL280)	55 authn/sec	65 authn/sec	81 authn/sec
Throughput including preauth (PL1200)	70 authn/sec	72 authn/sec	82 authn/sec
Preauth avg response time (PL60)	160 ms	60 ms	3 ms
Preauth avg response time (PL280)	100 ms	45 ms	3 ms
Preauth avg response time (PL1200)	100 ms	60 ms	3 ms
Login avg response time (PL60)	105 ms	100 ms	80 ms
Login avg response time (PL280)	120 ms	80 ms	65 ms
Login avg response time (PL1200)	110 ms	95 ms	95 ms

Notes:

- Caching public key and generating random data in AM increase overall performance, and also reduce the preauthentication response time as AM does not need to send public key and random generation requests to HSM.
- Higher PL has less authentication performance benefit because higher PL serves random generation and public key requests faster.
- HSM software version 2.3.0-368 has a slight performance improvement by 5% - 9%.

- 5% if the public key is not cached and random generation is done by HSM.
- 9% if the public key is cached and random generation is done by AM.

E2EEA Gemalto LunaEFT2 (Socket API, Firmware 2.4.0) - AuthenticationService - 1FA - E2EE Password

Note: E2EEA Gemalto LunaEFT2 via socket API is only available from AccessMatrix version 5.6.0.

Hardware

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	4 GB
DB - Oracle 12c Standard 2	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	6 GB
HSM	
HSM	Safenet LunaEFT2 PL1200 x1
Load Balancer	
Load Balancer	HAproxy hosted in the same machine as AM6

Configuration for tests

HSM firmware version	2.4.0
Password hash algo	SHA-256
RSA-OAEP hash algo for RSA-2048	SHA-256

Storage key (KTPV)	3DES, AES
Server random number length	16 bytes
AM connection pool size (1 HSM)	4
Number of realm accounts	1
Number of login accounts	1
Number of users in DB	6500K

Test Results

Public key	not cached		cached			
Random generation	HSM		HSM		AM	
Storage Key	3DES	AES	3DES	AES	3DES	AES
Throughput including preauth (PL1200)	75.2 authn/sec	78.8 authn/sec	85.4 authn/sec	82 authn/sec	92 authn/sec	92 authn/sec

E2EEA Gemalto PSE2 (PTK C) - AuthenticationService - 1FA - E2EE Password

Hardware

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	4 GB
DB - Oracle 12c Standard 2	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	6 GB
HSM	

AM6	
HSM	Safenet PSE2 PL25
HSM	Safenet PSE2 PL220

Configuration

Password hash algo	SHA-256
RSA-OAEP hash algo	SHA
Storage key (KTPV)	AES
Server random number length	16 bytes
Number of realm accounts	1
Number of login accounts	1
Number of users in DB	2200K

Test Scenario

- Preauthentication to generate E2EE session and to get a random number and public key.
- Login to verify user credential.

Test Result

RSA Key Size	**1024**	**2048**	**3072**	**4096**
Throughput (PL25)	52 authn/sec	25 authn/sec	8 authn/sec	8 authn/sec
Throughput (PL220)	180 authn/sec	75 authn/sec	30 authn/sec	15 authn/sec
Preauth avg response time (PL25)	150 ms	250 ms	110 ms	115 ms
Preauth avg response time (PL220)	45 ms	200 ms	250 ms	250 ms
Login avg response time (PL25)	220 ms	400 ms	920 ms	925 ms
Login avg response time (PL220)	300 ms	300 ms	370 ms	660 ms

Notes:

	- PL25 (1024-bit and higher), PL220 (2048-bit and higher): Bottleneck is at the HSM. AM6 and DB CPU usages are considered low, 30% or less. HSM usage level reaches 90+%. - PL220 (1024-bit): Bottleneck is at the DB. HSM CPU usage is about 85%.

E2EEA Thales PSE3 (PTK C) - AuthenticationService - 1FA - E2EE Password

Hardware

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	4 GB
DB - Oracle 12c Standard 2	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	6 GB
HSM	
HSM	Thales PSE3 PL25
HSM	Thales PSE3 PL220

Configuration

Password hash algo	SHA-256
RSA-OAEP hash algo	SHA
Storage key (KTPV)	AES
Server random number length	16 bytes
Number of realm accounts	1
Number of login accounts	1
Number of users in DB	1M

Test Scenario

- Preauthentication to generate E2EE session and to get a random number and public key.
- Login to verify user credential.

Test Result

E2EEA Thales PSE3 (PTK C) - AuthenticationService - 1FA - E2EE			
RSA Key Size	**2048**	**3072**	**4096**
Throughput (PSE3 PL25)	11.8 authn/sec	6 authn/sec	6 authn/sec
Throughput (PSE3 PL220)	100 authn/sec	52.5 authn/sec	52.5 authn/sec
Preauth avg response time (PL25)	1 ms	1 ms	1 ms
Preauth avg response time (PL220)	1 ms	1 ms	1 ms
Login avg response time (PL25)	840 ms	832 ms	829 ms
Login avg response time (PL220)	296 ms	375 ms	373 ms

Note: PL25, PL220: Bottleneck is at the HSM. AM6 and DB CPU usages are considered low, 30% or less. HSM usage level reaches 90+%. Preauth average response time is minified due to public key caching being enabled.

E2EEA Thales PSE3 (PTK C) - AuthenticationService - 1FA - E2EE Password(PQC)

Hardware

AM6	
Processor	Intel(R) Xeon(R) Silver 4416+
Cores/vCPU	2
Memory	4 GB
DB - MySQL v8.4.2	
Processor	Intel(R) Xeon(R) Silver 4416+
Cores/vCPU	2

AM6	
Memory	6 GB
HSM	
HSM	Thales PSE3 PL25

Configuration

Post Quantum Cryptography (PQC) Algorithm	ML_KEM_512 / ML_KEM_768 / ML_KEM_1024
Password hash algo	SHA-256
Wrapping key	AES
Number of users in DB	100M

Test Scenario

- Preauthentication to generate E2EE session and to get a random number and public key.
- Login to verify user credential.

Test Result

E2EEA Thales PSE3 (PTK C) - AuthenticationService - 1FA - E2EE(PQC)			
PQC Algorithm	ML_KEM_512	ML_KEM_768	ML_KEM_1024
Thread count	5	5	5
PreAuth Avg Response Time	1 ms	1 ms	1 ms
Login Avg Response Time	28 ms	34 ms	43 ms
AM CPU Usage	41 %	32 %	25 %
DB CPU Usage	34 %	28 %	22 %
HSM Usage	87 %	90 %	92 %
Throughput	163.6 authn/sec	135.1 authn/sec	109.7 authn/sec

E2EEA Utimaco CryptoServer - AuthenticationService - 1FA - E2EE Password

Hardware

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	2/4
Memory	2/4 GB
DB - Oracle 12c Standard 2	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4/12
Memory	6/16 GB
HSM	
HSM	Utimaco CryptoServer Se-Series Gen2 (Se1500/Se12/Se52)

Configuration

Password hash algo	SHA-256
RSA-OAEP hash algo	SHA
RSA Key Size	2048
Storage key (KTPV)	AES256
Server random number length	8 bytes
Number of realm accounts	1
Number of login accounts	1
Number of users in DB	1M

Test Scenario

- Preauthentication to generate E2EE session and to get a random number and public key.
- Login to verify user credential.

Test Result

AM6 Server	Database	HSM Firmware	No. of User Threads	Throughput	AM CPU usage	HSM CPU usage	DB CPU usage	Avg Response Time
4vCPU/4GB	4vCPU/6GB RAM	SE1500	30	300	80-90%	50-65%	50-65%	89ms
2vCPU/2GB	4vCPU/6GB RAM	SE1500	30	200	85-93%	20-40%	20-30%	138ms
4vCPU/4GB	12vCPU/16GB RAM	SE12	20	15	4-5%	99%	1-2%	1322ms
2vCPU/2GB	12vCPU/16GB RAM	SE12	20	15	9-13%	99%	1-2%	1322ms
4vCPU/4GB	12vCPU/16GB RAM	SE52	20	72	17-22%	99%	6-10%	272ms
2vCPU/2GB	12vCPU/16GB RAM	SE52	20	72	40-50%	99%	6-10%	272ms

Test Result with JWT Token

AM6 Server	Database	HSM Firmware	No. of User Threads	Throughput	AM CPU usage	HSM CPU usage	DB CPU usage	Avg Response Time
4vCPU/4GB	12vCPU/16GB RAM	SE12	20	15	8-10%	99%	1-2%	1318ms
2vCPU/2GB	12vCPU/16GB RAM	SE12	20	15	15-20%	99%	1-2%	1319ms
4vCPU/4GB	12vCPU/16GB RAM	SE52	20	72	35-45%	99%	8-13%	273ms
2vCPU/2GB	12vCPU/16GB RAM	SE52	20	72	72-82%	99%	8-13%	272ms

Notes:

- In the case of 4vCPU with SE1500 firmware, the AM CPU is reaching bottleneck. HSM and DB CPU usage is considered normal, 65% or less. AM CPU usage level is between 80-90%.

- In the case of 2vCPU with SE1500 firmware, the AM CPU is the bottleneck. HSM and DB CPU usage is considered low, 40% or less. AM CPU usage level reaches 90+%.
- In the case of 4vCPU with SE12 firmware, the HSM is the bottleneck. AM CPU and DB usage is considered low, 5% or less. HSM usage is at 99%. With JWT Token enabled, AM CPU usage increased to 10%.
- In the case of 2vCPU with SE12 firmware, the HSM is the bottleneck. AM CPU and DB usage is considered low, 10% or less. HSM usage is at 99%. With JWT Token enabled, AM CPU usage increased to 20%.
- In the case of 4vCPU with SE52 firmware, the HSM is the bottleneck. AM CPU and DB usage is considered low, 22% or less. HSM usage is at 99%. With JWT Token enabled, AM CPU usage increased to 45%.
- In the case of 2vCPU with SE52 firmware, the HSM is the bottleneck. AM CPU and DB usage is considered low, 50% or less. HSM usage is at 99%. With JWT Token enabled, AM CPU usage increased to 82%.

E2EEA Pseudo (non-HSM) - AuthenticationService - 1FA - E2EE Password

Hardware

Note: For Pseudo E2EEA, multiple stress tests were done over a period of time in slightly different environments (local VM server or in Google Cloud Platform). However, the hardware specs in each environment were generally the same. All were based on Intel Xeon E5 v3/v4, with a similar base CPU frequency.

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
vCPU	4/8/16
Memory	8 GB
Client	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
vCPU	16
Memory	64 GB

AM6	
Database	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
vCPU	8/16
Memory	16/64 GB

Configuration

Password hash algo	SHA-256
RSA-OAEP hash algo	SHA
Storage key (KTPV)	AES
RSA key pair is stored in	AM (either wrapped by HSM key or non-HSM key)
Storage key is stored in	AM (either wrapped by HSM key or non-HSM key)
Number of realm accounts	1
Number of login accounts	1
Number of users in DB	2500K

Test Scenario

All scenarios start with preauthentication, creation of RPIN in client-side, and reset/login (i.e preauthenticate -> create RPIN -> reset/login).

Stress test aims to have around 80% CPU usage on the AM server.

Test Result (vertical scaling)

Password reset:

Server	Throughput including preauth
4 vCPUs	220 reset/sec
8 vCPUs	335 reset/sec

Server	Throughput including preauth
16 vCPUs	530 reset/sec

Login:

Server	Throughput including preauth
4 vCPUs	250 authn/sec
8 vCPUs	430 authn/sec
16 vCPUs	760 authn/sec

Notes:

- DB is using 16 vCPU.
- When testing AM with a 16 vCPU, a two-client setup was used, wherein each client utilized no more than 100 threads. The following changes were also made:
    - Event manager worker thread count/max: 50/150 -> 200/600.
    - Datasource configuration in server: initialSize 10 -> 80, maxTotal 80 -> 200.

Test Result (horizontal scaling)

Login:

Server	Throughput including preauth
1 instance x 4 vCPU = 4 vCPU	250 authn/sec
3 instances x 4 vCPUs = 12 vCPU	710 authn/sec

Notes:

- DB is using 8 vCPU.
- 1 instance x 4 vCPU throughput is taken from "Test Result (vertical scaling)".
- For 3 instances x 4 vCPU, both DB and AM CPU usages are the bottleneck.

Test Scenario

- Preauthentication to generate E2EE session and to get a random number and public key.
- Login to verify user credential.

Test Result

AM6 Server	Database	No. of User Threads	Throughput	AM CPU usage	DB CPU usage	Avg Response Time
4vCPU/4GB	4vCPU/16GB RAM	10	242	58-62%	41-43%	34ms
4vCPU/4GB	4vCPU/16GB RAM	20	249	62-66%	51-57%	68ms

Test Result with JWT Token

AM6 Server	Database	No. of User Threads	Throughput	AM CPU usage	DB CPU usage	Avg Response Time
4vCPU/4GB	4vCPU/16GB RAM	10	148	65-68%	34%	60ms
4vCPU/4GB	4vCPU/16GB RAM	20	156	74-77%	46-48%	116ms

Note: DB is using 4 vCPU.

E2EEA Pseudo (non-HSM) - AuthenticationService - 1FA - PDF PIN Mailer

Hardware

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
vCPU	4
Memory	4 GB
Client	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
vCPU	16
Memory	64 GB

AM6	
Database	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
vCPU	4
Memory	16 GB

Configuration

Password hash algo	SHA-256
RSA-OAEP hash algo	SHA
Storage key (KTPV)	AES
RSA key pair is stored in	AM (either wrapped by HSM key or non-HSM key)
Storage key is stored in	AM (either wrapped by HSM key or non-HSM key)
Number of realm accounts	1
Number of login accounts	1
Number of users in DB	2500K

Test Scenario

1. Modify password/reset such that PDF mailer is generated but not delivered (e.g by commenting out the code that handles the delivery)
2. Perform stress test by making calls to this modified password/reset

Test Result

AM6 Server	No. of User Threads	Throughput	AM CPU usage	DB CPU usage	Avg Response Time
4vCPU/4GB	10	128.5	80-87%	19-22%	76ms
4vCPU/4GB	20	137.1	85-91%	20-25%	143ms

E2EEA Pseudo (Safenet PSE2 HSM) - AuthenticationService - 1FA - E2EE Password

HSM integration is used to directly encrypt the password hash using a key in HSM. RSA key pairs are in AM server.

Hardware

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	4 GB
DB - Oracle 12c Standard 2	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	6 GB
HSM	
HSM	Safenet PSE2 PL25
HSM	Safenet PSE2 PL220

Configuration

Password hash algo	SHA-256
RSA-OAEP hash algo	SHA
Storage key (KTPV)	AES
RSA key pair is stored in	AM (either wrapped by HSM key or non-HSM key)
Storage key is stored in	HSM
Server random number length	16 bytes

Number of realm accounts	1
Number of login accounts	1

Number of users in DB	2500K

Test Scenario

- Preauthentication to generate E2EE session and to get a random number and public key.
- Login to verify user credential.

Test Result

RSA Key Size	**2048**	**4096**
Throughput (PL25)	200 authn/sec	110 authn/sec
Throughput (PL220)	200 authn/sec	110 authn/sec

Note:

- AM CPU and DB CPU usages are the bottlenecks for RSA 2048-bit. For RSA 4096-bit, AM CPU usage is the bottleneck, whereas DB CPU usage is about 35%.
- HSM usage level for PL25 is below 30%, for PL220 is below 20%.

E2EE ATM PIN Translation - Thales payShield 10K

ATM Pin Translation in ISO Format 0.

Hardware

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	4 GB
DB - Oracle 12c Standard 2	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	6 GB
HSM	

AM6	
HSM	Thales payShield 10K

Test Scenario

- Preauthentication to generate E2EE session and to get a random number and public key.
- Client encrypts ISO-0 pin block with a public key.
- HSM translates RSA-encrypted pin block into TPK encrypted pin block.

Test Result

RSA Key Size	**2048**
Throughput	250 transaction/sec

Notes:

- AM CPU peaked at 25%.
- HSM CPU is relative low because its licence caps its performance to 250 transaction/sec. So we have already reached its max performance for the given license.

E2EEDP - End point on Application Server via AM6 Server (NEW)

Hardware

AM6	
Processor	Intel Xeon E5-2673 v4 @2.3GHz
vCPU	4/8/16
Memory	16/32/64 GB

Configuration

End point	Application Server via AM6 Server
Asymmetric Cipher:	
Number of Key Pairs	4
Key size	2048

End point	Application Server via AM6 Server
Symmetric Cipher:	
Algorithm	AES256
Integrity Check	
Algorithm	SHA256
Others	
Random number length	16

Test Scenario

For each request,

1. Client side generates a 16 byte string, call dataproc/params/get and E2EEDP client API to encrypt it.
2. Call dataproc/decrypt to decrypt string at AM6 server.

Stress test aims to have around 80% CPU usage on the AM server.

Test Result

Full scenario:

Server	Throughput
4 vCPUs / 16GB	204.25 tps
8 vCPUs / 32GB	409.45 tps
16 vCPUs / 64GB	824.6 tps

Call dataproc/params/get only:

Server	Throughput
4 vCPUs / 16GB	310.5 tps
8 vCPUs / 32GB	556.98 tps
16 vCPUs / 64GB	1055.68 tps

OATH - AuthenticationService - 2FA - Password and OATH verify OTP

Hardware

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	4 GB
DB - Oracle 12c Standard 2	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	6 GB

Configuration

```
spec1.setAlgorithm(TokenSpecTO.OATH_ALGO_TOTP);
spec1.setResponseFormatLength(6);
spec1.setTimeInterval(5);//secs
spec1.setSuite("HmacSHA256");
```

Password hash algo	SHA-256
Number of realm accounts	2
Number of login accounts	2
Number of token seeds/blobs per token	2

Test Result

	w/ audit w/ history			w/o audit w/o history		
Num of users in DB	10K	500K	1000K	10K	500K	1000K
Num of tokens in DB	10K	500K	1000K	10K	500K	1000K

	w/ audit w/ history			w/o audit w/o history		
Throughput	150 authn/sec	120 authn/sec	120 authn/sec	265 authn/sec	240 authn/sec	240 authn/sec

OATH - TokenService - OATH verify OTP

Hardware

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	4 GB
DB - Oracle 12c Standard 2	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	6 GB

Configuration

```
spec1.setAlgorithm(TokenSpecTO.OATH_ALGO_TOTP);
spec1.setResponseFormatLength(6);
spec1.setTimeInterval(5);//secs
spec1.setSuite("HmacSHA256");
```

Number of token seeds/blobs per token	2

Test Result

	w/ audit			w/o audit		
Num of users in DB	10K	500K	1000K	10K	500K	1000K
Num of tokens in DB	10K	500K	1000K	10K	500K	1000K

	w/ audit			w/o audit		
verifyOTP by UUID - Throughput	560 authn/sec	440 authn/sec	500 authn/sec	900 authn/sec	790 authn/sec	800 authn/sec
verifyOTP by Serial - Throughput	480 authn/sec	330 authn/sec	350 authn/sec	(not tested)	(not tested)	(not tested)

OATH - HSM - Import Token

Hardware

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	4 GB
DB - Oracle 12c Standard 2	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	6 GB
HSM	
HSM	Safenet PSE2 PL220 with i-Sprint FM 1.06

Configuration

Number of token seeds/blobs per token	4

Crypto operations involved in FM:

1. Decrypt encrypted transport key using KEK.
2. Decrypt encrypted seed using the transport key.
3. Encrypt seed using storage key.

Test Result

Token import of 10K tokens took about 6 minutes with very low AM and DB CPU usage. About 45% HSM usage level.
Token import of 30K tokens took about 14 minutes with very low AM and DB CPU usage. About 45% HSM usage level.
Token import of 50K tokens took about 23 minutes with very low AM and DB CPU usage. About 45% HSM usage level.

OATH - HSM - TokenService - OATH verify OTP

Hardware

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	4 GB
DB - Oracle 12c Standard 2	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	6 GB
HSM	
HSM	Safenet PSE2 PL220 with i-Sprint FM 1.06

Configuration

```
spec1.setAlgorithm(TokenSpecTO.OATH_ALGO_TOTP);
spec1.setResponseFormatLength(8);
spec1.setTimeInterval(30);//secs
spec1.setSuite("HmacSHA256");
```

Number of token seeds/blobs per token	4
Num of users in DB	1000K

Num of tokens in DB	1000K

Test Result

HSM usage level is around 45%. The bottleneck is in DB I/O.

	w/ audit	w/o audit
verifyOTP by UUID - Throughput	300 authn/sec	x
verifyOTP by Serial - Throughput	240 authn/sec	x

FIDO UAF - TokenService.UAF - Registration

Hardware

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	4 GB
DB - Oracle 12c Standard 2	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	6 GB
Client	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	8
Memory	4 GB

Configuration

Authenticator metadata (partial)
"authenticationAlgorithm": 3, "publicKeyAlgAndEncoding": 258, "attestationTypes": [15880],

Public key	RSA-2048
Authentication Algorithm	UAF_ALG_SIGN_RSASSA_PSS_SHA256_RAW
Public Key Algo and Encoding	UAF_ALG_KEY_RSA_2048_PSS_RAW
Attestation Type	TAG_ATTESTATION_BASIC_SURROGATE

Test Result

Heavy on DB, 60% CPU usage on DB, 30% CPU usage on AM.

uafRequest and uafProcess

	w/ audit	**w/o audit**
Throughput	35 authn/sec	\

FIDO UAF - TokenService.UAF - Authentication

Hardware

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	4 GB
DB - Oracle 12c Standard 2	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	6 GB
Client	

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	8
Memory	4 GB

Configuration

Authenticator metadata (partial)
"authenticationAlgorithm": 3, "publicKeyAlgAndEncoding": 258, "attestationTypes": [15880],

Public key	RSA-2048
Authentication Algorithm	UAF_ALG_SIGN_RSASSA_PSS_SHA256_RAW
Public Key Algo and Encoding	UAF_ALG_KEY_RSA_2048_PSS_RAW
Attestation Type	TAG_ATTESTATION_BASIC_SURROGATE

Test Result

With audit: Heavy on AM, 70% CPU usage on AM, 60% CPU usage on DB, 70% CPU usage on Client.

Without audit: Heavy on AM, 70% CPU usage on AM, 40% CPU usage on DB, 80% CPU usage on Client.

uafRequest and uafProcess

	w/ audit	w/o audit
Throughput	170 authn/sec	200 authn/sec
uafRequest response time	55 ms	240 ms
uafProcess response time	180 ms	390 ms
DB Read I/O	5 requests/sec	<1 request/sec
DB Write I/O	1100 requests/sec	230 requests/sec

VASCO Multi-Device - Import Token

Hardware

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	4 GB
DB - Oracle 12c Standard 2	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	6 GB

Configuration

Number of token seeds/blobs per token	1 (only MA Blob)

Test Result

Token import of 20K tokens took about 6 minutes with very low AM and DB CPU usage.

Token import of 30K tokens took about 11 minutes with very low AM and DB CPU usage.

VASCO Multi-Device - TokenService - Activation

3-step activation (license, instance, and post) using the multiDeviceActivationStep1(), multiDeviceActivationStep2(), and multiDeviceActivationStep3() APIs.

Hardware

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	4 GB

AM6	
DB - Oracle 12c Standard 2	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	6 GB
Client - Jmeter	
Processor	Intel Core i7-3770 @3.4GHz
Cores/vCPU	4
Memory	16 GB

Test Result

With audit: Heavy on AM, 90% CPU usage on AM, 50% CPU usage on DB, 20% CPU usage on Client.
Without audit: Heavy on AM and DB, 80% CPU usage on AM, 80% CPU usage on DB, 20% CPU usage on Client.

	w/ audit	w/o audit
Number of tokens in DB	100K	100K
Number of instance per token in DB	2	2
Number of token seeds/blobs per instance	4	4
Multi-steps Activation - Throughput	30 activations/sec	51 activations/sec
DB Read I/O	20 requests/sec	16 requests/sec
DB Write I/O	2000 requests/sec	1100 requests/sec

VASCO Multi-Device - TokenService - Activation(Redis Cluster)

3-step Activation (license, instance, and post) using the multiDeviceActivationStep1(), multiDeviceActivationStep2(), and multiDeviceActivationStep3() APIs with Redis cluster.

Hardware

AM6 - Amazon EC2, M5 series	
Processor	Intel Xeon® Platinum 8175M
Cores/vCPU	4 to 16 GB
Memory	16 to 64 GB
DB - Amazon RDS MySQL, R5 series	
Processor	Intel Xeon Platinum 8000
Cores/vCPU	8 to 32 GB
Memory	64 to 256 GB
Redis cache	
Engine	Redis 7.1.0
Cores/vCPU	2
Memory	0.5 GB
Cluster mode	Enabled
Multi-AZ	Enabled
Shards	3
No. of nodes	6

Test result

Instances	AM6 Server	Database	AM6 CPU usage	DB CPU usage	Threads	Throughput(activations/sec)	Avg response time(ms)
1	4vCPU/16GB	8vCPU/64GB	70%	70%	10000	15	116

Instances	AM6 Server	Database	AM6 CPU usage	DB CPU usage	Threads	Throughput(activations/sec)	Avg response time(ms)
2	4vCPU/16GB	8vCPU/64GB	70%	60%	20000	33	99
2	16vCPU/64GB	32vCPU/256GB	80%	50%	45000	149	97
3	16vCPU/64GB	32vCPU/256GB	80%	80%	45000	218	288

VASCO Multi-Device - TokenService - verify OTP

OTP verification using the verifyOTPBySerialNum() API.

Hardware

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	4 GB
DB - Oracle 12c Standard 2	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	6 GB
Client - Jmeter	
Processor	Intel Core i7-3770 @3.4GHz
Cores/vCPU	4
Memory	16 GB

Test Result

With audit: Heavy on DB, 90% CPU usage on DB, 50% CPU usage on AM, 20% CPU usage on Client.

Without audit: Average on AM and DB, 50% CPU usage on AM, 60% CPU usage on DB, 20% CPU usage on Client.

	w/ audit		w/o audit	
Number of tokens in DB	100K	100K	100K	100K
Number of instance per token	1	3	1	3
verifyOTP by Serial - Throughput	200 authn/sec	180 authn/sec	530 authn/sec	430 authn/sec
DB Read I/O	16 requests/sec	56 requests/sec	<1 request/sec	2 requests/sec
DB Write I/O	1700 requests/sec	2100 requests/sec	700 requests/sec	756 requests/sec

VASCO Multi-Device - TokenService - verify OTP (REST)

OTP verification using the token/verifyOTP REST API.

Hardware

AM6 - Amazon EC2 M5 Series	
Processor	Intel(R) Xeon(R) Platinum 8175M CPU @ 2.50GHz
Cores/vCPU	4 to 16
Memory	16 to 64 GB
DB - Amazon Aurora DB for MySQL 2.10.1, R5 Series	
Processor	Intel(R) Xeon(R) Platinum 8175M CPU @ 2.50GHz
Cores/vCPU	8 to 32
Memory	32 to 128 GB
Client - Jmeter on Amazon EC2 M5 Series	
Processor	Intel(R) Xeon(R) Platinum 8175M CPU @ 2.50GHz

AM6 - Amazon EC2 M5 Series	
Cores/vCPU	4 to 16
Memory	16 to 64 GB

Test Result

1x 4vCPU/16GB AM instance on 4vCPU/16GB Aurora DB	
Throughput	209 verifications/sec
AM CPU usage	80%
DB CPU usage	65-70%
3x 8vCPU/32GB AM instances on 16vCPU/64GB Aurora DB	
Throughput	1200 verifications/sec
AM CPU usage	80%
DB CPU usage	90%
5x 8vCPU/32GB AM instances on 32vCPU/128GB Aurora DB	
Throughput	2000 verifications/sec
AM CPU usage	85%
DB CPU usage	75%
5x 8vCPU/32GB AM instances on 32vCPU/128GB Aurora DB w/o Audit	
Throughput	3000 verifications/sec
AM CPU usage	88%
DB CPU usage	70%
4x 16vCPU/64GB AM instances on 32vCPU/128GB Aurora DB	
Throughput	2400 verifications/sec
AM CPU usage	60-65%
DB CPU usage	85%

1x 4vCPU/16GB AM instance on 4vCPU/16GB Aurora DB	
4x 16vCPU/64GB AM instances on 32vCPU/128GB Aurora DB w/o Audit	
Throughput	3800 verifications/sec
AM CPU usage	65%
DB CPU usage	95%

Additional Information

Test Setup	No. of User Threads	DB Connection Pool Size	AM Event Manager Worker Pool Size
1x 4vCPU/16GB AM instance on 4vCPU/16GB Aurora DB	40	80(default)	50(default)
3x 8vCPU/32GB AM instances on 16vCPU/64GB Aurora DB	50	160	100
5x 8vCPU/32GB AM instances on 32vCPU/128GB Aurora DB	50 70 w/o audit	200	100
4x 16vCPU/64GB AM instances on 32vCPU/128GB Aurora DB	90 125 w/o audit	320	100

VASCO Multi-Device - TokenService - verifyMessageSignature

For Secure Channel Transaction Signing. The test includes generating Secure Channel Message and response verification.

APIs used: generateSecureChannelMessageRequest(), verifyMessageSignature()

Hardware

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4

AM6	
Memory	4 GB
DB - Oracle 12c Standard 2	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	6 GB
Client - Jmeter	
Processor	Intel Core i7-3770 @3.4GHz
Cores/vCPU	4
Memory	16 GB

Test Result

With audit: Heavy on DB, 90% CPU usage on DB, 60% CPU usage on AM, 20% CPU usage on Client.

Without audit: Heavy on AM, 80% CPU usage on AM, 40% CPU usage on DB, 20% CPU usage on Client.

	w/ audit		w/o audit	
Number of tokens in DB	100K	100K	100K	100K
Number of instance per token	1	3	1	3
Generate & Verify - Throughput	119 activations/sec	90 activations/sec	222 activations/sec	133 activations/sec
DB Read I/O	30 requests/sec	21 requests/sec	<1 request/sec	<1 request/sec
DB Write I/O	1800 requests/sec	1250 requests/sec	250 requests/sec	200 requests/sec

VASCO Multi-Device - TokenService - verifyMessageSignature (REST)

For Secure Channel Transaction Signing. The test includes generating Secure Channel Message and response verification.

APIs used: VASCO/multidev/message/secureChannel/generateRequest, VASCO/multidev/message/signature/verify

Hardware

AM6 - Amazon EC2, M5 Series	
Processor	Intel(R) Xeon(R) Platinum 8175M CPU @ 2.50GHz
Cores/vCPU	4 to 16 vCPU
Memory	16 to 64 GB
DB - Amazon RDS MySQL, R5 Series	
Processor	Intel(R) Xeon(R) Platinum 8000 CPU @ 3.1GHz
Cores/vCPU	4 to 32 vCPU
Memory	32 to 256 GB
Client - Jmeter	
Processor	Intel(R) Xeon(R) Platinum 8175M CPU @ 2.50GHz
Cores/vCPU	4 to 16 vCPU
Memory	16 to 64 GB

Test Result

Instances	AM6	DB	Client	User Threads	Throughput	Avg Resp Time	AM6 CPU %	DB CPU %	Client CPU %
1	4vCPU	4vCPU	4vCPU	30	136.9 verifications/sec	102 ms	92%	95%	17%
1	16vCPU	32vCPU	16vCPU	60	454.7 verifications/sec	62 ms	92%	36%	13%

Instances	AM6	DB	Client	User Threads	Throughput	Avg Resp Time	AM6 CPU %	DB CPU %	Client CPU %
2	16vCPU	32vCPU	16vCPU	120	832.1 verifications/sec	64 ms	92%	51%	28%
3	16vCPU	32vCPU	16vCPU	300	1173.8 verifications/sec	95 ms	77%	58%	40%

VASCO Multi-Device - TokenService - verifyDigitalSignature

For Transaction Signing without Secure Channel. The test includes generating and verifying digital signature.

APIs used: token/verifyDigitalSignature

Hardware

AM6 - Amazon EC2, M5 Series	
Processor	Intel(R) Xeon(R) Platinum 8175M CPU @ 2.50GHz
Cores/vCPU	4 to 16 vCPU
Memory	16 to 64 GB
DB - Amazon RDS MySQL, R5 Series	
Processor	Intel(R) Xeon(R) Platinum 8000 CPU @ 3.1GHz
Cores/vCPU	4 to 32 vCPU
Memory	32 to 256 GB
Client - Jmeter	
Processor	Intel(R) Xeon(R) Platinum 8175M CPU @ 2.50GHz

AM6 - Amazon EC2, M5 Series	
Cores/vCPU	4 to 16 vCPU
Memory	16 to 64 GB

Test Result

Instances	AM6	DB	Client	User Threads	Throughput	Avg Resp Time	AM6 CPU %	DB CPU %	Client CPU %
1	4vCPU	4vCPU	4vCPU	20	207 verifications/sec	88 ms	92%	88%	27%
1	16vCPU	32vCPU	16vCPU	50	758.2 verifications/sec	51 ms	83%	33%	26%
2	16vCPU	32vCPU	16vCPU	100	1188.8 verifications/sec	60 ms	88%	30%	49%
3	16vCPU	32vCPU	16vCPU	250	1559 verifications/sec	109 ms	90%	28%	64%

SMSOTP Token - TokenService - Verify OTP (REST)

The test includes generating and verifying the OTP.

Hardware

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4vCPU
Memory	4 GB
DB - Oracle 12c Standard 2	
Processor	Intel Xeon E5-2620 v3 @2.4GHz

AM6	
Cores/vCPU	4vCPU
Memory	16 GB

Test Scenario

For each request,

- /token/generateOTP is first called to generate the OTP and sent via HTTP Deliverer.
- /token/verifyOTP is then called to verify the OTP generated.

Test Result

	w/ audit	w/o audit
Throughput (verifyOTP/sec)	215	863
Avg resp time(ms/verifyOTP)	45	10
AM CPU Usage	40-60	85-95
DB CPU Usage	75-95	75-85

Note: When the delivery channel and user input is longer the bottleneck, the SMSOTP process is:

- DB focused when audit is enabled
- Server focused when audit is disabled

SenseTime Face Biometric - BiometricService - enrollTemplates

Biometric template enrollment using enrollTemplates() API.

In addition to AM memory requirement, SenseTime Face Biometric requires additional 4GB for an object pool size of 10 (default), which is suitable for 8vCPU.

Hardware

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	8vCPU
Memory	12 GB
DB - Oracle 12c Standard 2	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4vCPU
Memory	6 GB
Client - Jmeter	
Processor	Intel Core i7-3770 @3.4GHz
Cores	4C8T
Memory	16 GB

Software

SenseTime	
SenseTime FaceSDK	v6.4.3 Linux 64-Bit
SenseTime AntihackSDK	v5.2.1 Linux 64-Bit
SenseTime Object Pool Created Count	10 For FaceSDK, 10 For AntihackSDK

Test Result

With audit: Heavy on AM, 95% CPU usage on AM. Most of the CPU resources were used for face feature extraction by SenseTime FaceSDK. Very low DB CPU usage at around 5%.

	w/ audit	
With Antihack	**Yes**	**No**
Number of users in DB	2000K	2000K
Number of users enrolled	200K	200K

	w/ audit	
Input Image Size	449 KB	449 KB
enrollTemplates - Throughput	11 enrolments/sec	15 enrolments/sec
DB Read I/O	50 requests/sec	50 requests/sec
DB Write I/O	70 requests/sec	70 requests/sec

SenseTime Face Biometric - BiometricService - verifyTemplates

Biometric template verification using verifyTemplates() API.

In addition to AM memory requirement, SenseTime Face Biometric requires additional 4GB (6GB with anti-hack enabled) for an object pool size of 10 (default), which is suitable for 8vCPU.

Hardware

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	8vCPU
Memory	12 GB
DB - Oracle 12c Standard 2	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4vCPU
Memory	6 GB
Client - Jmeter	
Processor	Intel Core i7-3770 @3.4GHz
Cores	4C8T
Memory	16 GB

Software

SenseTime	
SenseTime FaceSDK	v6.4.3 Linux 64-Bit
SenseTime AntihackSDK	v5.2.1 Linux 64-Bit
SenseTime Object Pool Created Count	10 For FaceSDK, 10 for Antihack SDK

Test Result

With audit: Heavy on AM, 95% CPU usage on AM. Most of the CPU resources were used for face feature extraction by SenseTime FaceSDK. Very low DB CPU usage at around 2-5%.

	w/ audit	
With Antihack	**Yes**	**No**
Number of users in DB	2000K	2000K
Number of users with templates in DB	200K	200K
Input Image Size	449 KB	449 KB
verifyTemplates - Throughput	10 verifications/sec	15 verifications/sec
DB Read I/O	<30 requests/sec	<30 requests/sec
DB Write I/O	<30 requests/sec	<30 requests/sec

YESsafe (SafeNet PSE Network HSM) - Token Activation

Hardware

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	4GB
DB	
Processor	Intel Xeon E5-2620 v3 @2.4GHz

AM6	
Cores/vCPU	4
Memory	16GB
HSM	
HSM	Safenet PSE2 PL220 with i-Sprint FM "E2EEFM 1.13 V1.13"

Configuration

Num of users in DB	100K
Num of tokens in DB	100K

Test Result

Number of user threads: 25

AM CPU usage: around 80%

DB CPU usage: 70-80%

	Throughput
OATH/create-assign-and-encrypt	51.4/sec
token/activateByOTP	134.5/sec
device/find, device/setRooted	134.5/sec

Note: device/notification/push/register is not implemented in the HSM OATH Token Handler.

YESsafe (non-HSM) - Token Activation

Hardware

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4

AM6	
Memory	4GB
DB	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	16GB

Configuration

Num of users in DB	100K
Num of tokens in DB	100K

Test Result

Number of user threads: 25

AM CPU usage: 80-90%

DB CPU usage: 70-80%

	Throughput
OATH/create-assign-and-encrypt	53.3/sec
token/activateByOTP	81.6/sec
device/find, device/notification/push/register, device/setRooted	82.3/sec

Note: Without calling device/notification/push/register, the throughput of token activation will be increased to 118.3/sec.

FIDO2 - Token Service - assertionFinish

Hardware

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz

AM6	
Cores/vCPU	4
Memory	4GB
DB	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	16GB

Test Result

	DB CPU Usage	AM CPU Usage	Throughput
30 Threads	75%	85%	206/sec
40 Threads	86%	90%	212/sec

Notes:

- FIDO2 has 4 processes, namely registration start/finish and assertion start/finish. This stress test only covers assertion finish.
- The other 3 processes were run on a client in a separate machine to generate the user's credential before starting this test.

2.2.1. OpenShift/Kubernetes Scalability

This section presents the stress test throughput results for scaling in OpenShift/Kubernetes. The data presented here will be useful for resource planning.

OpenShift is a container platform built on Kubernetes by Red Hat. It is designed for faster, more efficient application deployment. As businesses modernize their IT infrastructure, many are shifting from virtual machines (VM) to OpenShift to take advantage of its speed, scalability, and resource efficiency. Unlike VMs, which require a full operating system for each instance, containers share the OS kernel, reducing overhead and improving performance. This allows applications to deploy in seconds rather than minutes and scale dynamically without provisioning new VMs. With seamless portability across hybrid and multi-cloud environments, OpenShift is becoming the preferred choice for enterprises looking to enhance agility and reduce costs.

Disclaimer

This performance scaling report provides reference throughput based on the number of pods, vCPUs, and RAM. However, actual performance may vary due to several factors beyond these numbers, such as processor generation, clock speed, cache size, RAM type and speed, and storage performance. e.g. A newer processer generation may potentially have higher performance given the same number of cores and same clock speed. It is recommended to factor these as part of your resource planning.

Database tuning is equally important. Increasing the vCPU and memory of the database server is often not sufficient to handle the increase in traffic, when increasing the number of AccessMatrix pods . Database settings such as I/O throughput need to be increased as well. This is beyond the scope of this guide. For cloud managed databases (such as those provided by AWS or Azure), the database configuration is already optimized for the selected database vCPU/memory size. For self-managed databases, it is recommended to use the same settings as the baseline for further tuning.

Hardware

A dedicated OpenShift worker node was used for the performance test. OpenShift background and management related applications are dedicated to a separate node. This is to ensure that the stress test results will not be affected by OpenShift background processes. The specification of the dedicated worker node is given below.

OpenShift Dedicated Worker Node	
Processor	Intel(R) Xeon(R) Silver 4416+ CPU @ 2.00GHz
Cores/vCPU	80
Memory	256 GB

Scaling of Kubernetes pods

In OpenShift/Kubernetes, throughput scaling is achieved by increasing/decreasing the number of pods (a.k.a replicas) during high/low load. The choice for the number of vCPU and RAM assigned to each pod depends on the usage pattern. A pod with say 2 vCPU and 4 GB RAM would be resource efficient, but would also reach its throughput limit faster than say a pod with 8 vCPU and 16 GB RAM. In order to suit the needs of different customers, performance test is conducted for 4 pod profiles and a total of 9 test configurations as shown below.

Pod Profile	**No. of pods**	**Total (AccessMatrix)**		**DB (vCPU/RAM)**	
		vCPU	**RAM (GB)**	**vCPU**	**RAM (GB)**
micro (2vCPU/4GB RAM per pod)	1	2	4	2	6
	2	4	8	4	12
	6	12	24	12	24
small (4vCPU/8GB RAM per pod)	2	8	16	8	16
	4	16	32	16	32
	6	24	48	24	48
medium (8vCPU/16GB RAM per pod)	2	16	32	16	32
	4	32	64	32	64
large (16v CPU/32GB RAM per pod)	2	32	64	32	64

Note: For the medium profile, 6 pods is not tested. For the large profile, 4 pods and 6 pods are not tested. This is because the hardware has run out

	of vCPU cores. Further addition of pods will result in CPU contention, which will not represent the true performance. It is part of our roadmap to continue to increase the hardware resources for stress testing. Additional results will be added once available.

User size and token size for stress testing

For each of the 9 test configurations above, the stress test is repeated for 20 million users/tokens, and 100 million users/tokens. This is to identify performance dips when the number of users and tokens increases.

Profile	No. of users in database	No. of tokens in database
20M	20 million	20 million
100M	100 million	100 million

Test Cases

Use Case	Category	Variations	Test Case Name	REST API Endpoints
Password	default	-	SystemPasswordBasic	authn/login
	RSA Keys	Key length = 2048	PseudoE2EEA_RSA2048	authn/login
		Key length = 1024	PseudoE2EEA_RSA4096	authn/login
	Post-Quantum Safe keys	Key strength = 512	PseudoE2EEA_PQC512	authn/login
		Key strength = 768	PseudoE2EEA_PQC768	authn/login
		Key strength = 1024	PseudoE2EEA_PQC1024	authn/login
OneSpan Token	Activation	Step 1	OneSpan_Activation_Step1	token/VASCO/multidev/activate/step/1
		Step 2	OneSpan_Activation_Step2	token/VASCO/multidev/activate/step/2
		Step 3	OneSpan_Activation_Step3	token/VASCO/multidev/activate/step/3
	Token APIs	OTP	OneSpan_Token_RO	token/verifyOTP
		Transaction Signing	OneSpan_Token_SG	token/verifyDigitalSignature
		Secure Channel Transaction Signing	OneSpan_Token_SC	VASCO/multidev/message/signature/verify

Use Case	Category	Variations	Test Case Name	REST API Endpoints
	Login APIs	OTP	OneSpan_Login_RO	authn/login
		Transaction Signing	OneSpan_Login_SG	authn/login
		Secure Channel Transaction Signing	OneSpan_Login_SC	authn/login
YESsafe Token	Provisioning	Create and Assign	OATH_Create_Assign	token/OATH/create-assign-and-encrypt
		Activation	OATH_Activation	token/activateByOTP
	Token APIs	TOTP	OATH_Token_TOTP	token/verifyOTP
		OCRA	OATH_Token_OCRA	token/verifyResponse
	Login APIs	TOTP	OATH_Login_TOTP	authn/login
		OCRA	OATH_Login_OCRA	authn/login

Results

Throughput Per Second (TPS) for 20M

Profile	**micro**			**small**			**medium**		**large**
No. of pods	1	2	6	2	4	6	2	4	2
Total vCPU	2	4	12	8	16	24	16	32	32
Test Case Name	**TPS for 20M**								
SystemPasswordBasic	219	338	1475	1082	1770	2178	1735	2522	2498
PseudoE2EEA_RSA2048	215	415	1040	757	1295	1656	1312	1975	1986
PseudoE2EEA_RSA4096	134	248	654	470	848	1120	847	1355	1378
PseudoE2EEA_PQC512	240	453	1123	830	1406	1787	1526	2110	2107
PseudoE2EEA_PQC768	230	447	1104	808	1380	1752	1513	2089	2076
PseudoE2EEA_PQC1024	229	421	1037	767	1300	1679	1404	2037	1988
OneSpan_Activation_Step1	17	22	53	39	62	69	56	65	64
OneSpan_Activation_Step2	17	22	53	39	62	69	56	65	64
OneSpan_Activation_Step3	17	22	53	39	62	70	57	65	64

OneSpan_Token_RO	117	231	570	480	685	791	709	921	888
OneSpan_Token_SG	101	193	600	418	754	940	772	1092	1139
OneSpan_Token_SC	65	152	427	296	542	701	542	839	866
OneSpan_Login_RO	34	67	172	125	225	306	224	378	378
OneSpan_Login_SG	47	93	239	173	302	383	329	462	478
OneSpan_Login_SC	40	82	207	149	260	334	279	411	408
OATH_Create_Assign	50	68	207	180	252	266	262	279	273
OATH_Activation	100	240	602	415	751	957	824	1176	1143
OATH_Token_TOTP	180	331	946	779	1198	1518	1310	1924	1959
OATH_Token_OCRA	174	322	934	784	1136	1471	1178	1907	1900
OATH_Login_TOTP	71	152	371	276	459	570	491	682	689
OATH_Login_OCRA	68	137	319	241	396	464	438	516	506

Throughput Per Second (TPS) for 100M

Profile	**micro**			**small**			**medium**		**large**
No. of pods	1	2	6	2	4	6	2	4	2
Total vCPU	2	4	12	8	16	24	16	32	32
Test Case Name	**TPS 100M**								
SystemPasswordBasic	210	328	1369	1009	1625	2126	1635	2463	2410
PseudoE2EEA_RSA2048	210	401	980	725	1230	1638	1226	1965	1961
PseudoE2EEA_RSA4096	131	251	637	461	821	1133	838	1357	1381
PseudoE2EEA_PQC512	234	420	1012	758	1296	1731	1402	2100	2046
PseudoE2EEA_PQC768	228	428	1026	756	1284	1711	1412	2082	2051
PseudoE2EEA_PQC1024	218	399	952	706	1241	1567	1352	1911	1845
OneSpan_Activation_Step1	17	23	52	37	61	67	55	67	65
OneSpan_Activation_Step2	17	23	52	37	61	67	55	67	65
OneSpan_Activation_Step3	17	23	53	38	62	67	56	67	65
OneSpan_Token_RO	107	224	619	455	637	759	676	839	853

OneSpan_Token_SG	103	230	559	375	724	886	782	1030	994
OneSpan_Token_SC	71	171	400	289	535	697	550	794	823
OneSpan_Login_RO	34	64	163	120	211	280	209	357	350
OneSpan_Login_SG	46	91	221	170	288	380	300	350	456
OneSpan_Login_SC	39	76	189	148	238	330	252	324	377
OATH_Create_Assign	46	60	187	163	229	282	228	258	267
OATH_Activation	98	232	623	499	737	916	822	1137	1178
OATH_Token_TOTP	240	312	995	763	1254	1524	1330	1888	1856
OATH_Token_OCRA	205	304	943	750	1187	1445	1206	1895	1829
OATH_Login_TOTP	73	136	320	240	394	546	428	651	654
OATH_Login_OCRA	67	124	300	225	357	440	411	460	477

Percentage (%) change in TPS when users/tokens increased from 20M to 100M

On average, the TPS will drop by 3.3% when the number of users/tokens have been increased from 20M to 100M (5x increase).

Profile	**micro**			**small**			**medium**		**large**	**Avg.**
No. of pods	1	2	6	2	4	6	2	4	2	
Total vCPU	2	4	12	8	16	24	16	32	32	
Test Case Name	**% TPS change from 20M to 100M (-ve means drop in TPS)**									
SystemPasswordBasic	-4.2%	-2.9%	-7.2%	-6.7%	-8.2%	-2.4%	-5.7%	-2.3%	-3.5%	**-4.8%**
PseudoE2EEA_RSA2048	-2.5%	-3.4%	-5.8%	-4.2%	-5.1%	-1.1%	-6.5%	-0.5%	-1.3%	**-3.4%**
PseudoE2EEA_RSA4096	-1.8%	1.1%	-2.6%	-1.9%	-3.3%	1.2%	-1.1%	0.2%	0.2%	**-0.9%**
PseudoE2EEA_PQC512	-2.5%	-7.2%	-9.9%	-8.7%	-7.9%	-3.1%	-8.1%	-0.5%	-2.9%	**-5.6%**
PseudoE2EEA_PQC768	-1.1%	-4.3%	-7.0%	-6.4%	-7.0%	-2.4%	-6.7%	-0.3%	-1.2%	**-4.0%**
PseudoE2EEA_PQC1024	-5.1%	-5.2%	-8.2%	-8.0%	-4.6%	-6.7%	-3.8%	-6.2%	-7.2%	**-6.1%**

OneSpan_Activation_Step1	0.0%	0.9%	-1.3%	-3.6%	-0.5%	-3.6%	-2.0%	2.5%	1.3%	**-0.7%**
OneSpan_Activation_Step2	-0.6%	0.9%	-1.5%	-3.9%	-0.5%	-3.7%	-1.8%	2.5%	1.3%	**-0.8%**
OneSpan_Activation_Step3	0.0%	0.9%	-1.1%	-3.6%	-0.3%	-3.6%	-1.8%	2.6%	1.4%	**-0.6%**
OneSpan_Token_RO	-9.2%	-2.9%	8.6%	-5.2%	-6.9%	-4.0%	-4.7%	-9.0%	-4.0%	**-4.1%**
OneSpan_Token_SG	1.4%	18.8%	-6.7%	-10.2%	-3.9%	-5.8%	1.3%	-5.7%	-12.8%	**-2.6%**
OneSpan_Token_SC	8.6%	12.6%	-6.3%	-2.2%	-1.2%	-0.6%	1.5%	-5.4%	-4.9%	**0.2%**
OneSpan_Login_RO	1.2%	-4.3%	-5.2%	-3.8%	-6.3%	-8.3%	-6.8%	-5.6%	-7.4%	**-5.2%**
OneSpan_Login_SG	-1.1%	-2.5%	-7.7%	-1.4%	-4.7%	-0.8%	-8.7%	-24.2%	-4.7%	**-6.2%**
OneSpan_Login_SC	-3.5%	-7.5%	-8.8%	-1.1%	-8.4%	-1.1%	-9.7%	-21.3%	-7.6%	**-7.7%**
OATH_Create_Assign	-7.2%	-12.6%	-9.4%	-9.4%	-9.0%	6.2%	-13.2%	-7.3%	-2.1%	**-7.1%**
OATH_Activation	-2.4%	-3.2%	3.5%	20.3%	-1.9%	-4.3%	-0.3%	-3.3%	3.0%	**1.3%**
OATH_Token_TOTP	32.9%	-5.8%	5.1%	-2.1%	4.7%	0.4%	1.5%	-1.9%	-5.2%	**3.3%**
OATH_Token_OCRA	18.0%	-5.8%	0.9%	-4.3%	4.4%	-1.7%	2.4%	-0.6%	-3.7%	**1.1%**
OATH_Login_TOTP	1.7%	-10.7%	-13.7%	-12.8%	-14.2%	-4.3%	-12.8%	-4.5%	-5.1%	**-8.5%**
OATH_Login_OCRA	-0.9%	-9.8%	-5.9%	-6.6%	-9.9%	-5.1%	-6.1%	-10.9%	-5.6%	**-6.7%**
Average	**1.0%**	**-2.5%**	**-4.3%**	**-4.1%**	**-4.5%**	**-2.6%**	**-4.4%**	**-4.8%**	**-3.4%**	**-3.3%**

2.3. UAM

Session - Refresh

This test refreshes valid session tokens of authenticated users by calling the session/refresh API.

Hardware

AM6	
Processor	Intel Xeon® Platinum 8175M @3.1GHz
Cores/vCPU	4
Memory	16 GB
DB - Oracle 12c Standard 2	
Processor	Arm-based AWS Graviton2
Cores/vCPU	4
Memory	32 GB

Configuration

Number of valid sessionTokens	50,000

Test Result

	w/o Session Persistence	w/ Session Persistence
Num of sessionTokens in CSV	50,000	50,000
Throughput	846.8 requests/sec	748.0 requests/sec
Avg Resp Time	12 ms	20 ms
AM CPU Usage	Around 90%	Around 90%
DB CPU Usage	Around 56%	Around 73%

Session - Verify

Test to verify user session by calling the session/verify API.

Hardware

AM6	
Processor	Intel Xeon® Platinum 8175M @3.1GHz
Cores/vCPU	4
Memory	16 GB
DB - Aurora MySQL 8.0	
Processor	Arm-based AWS Graviton2
Cores/vCPU	4
Memory	32 GB

Configuration

Number of valid sessionToken	50,000

Test result

	w/o Session Persistence	w/ Session Persistence
Num of sessionTokens in CSV	50,000	50,000
Throughput	1044.8 requests/sec	1006.2 requests/sec
Avg Resp Time	8 ms	11 ms
AM CPU Usage	Around 90%	Around 90%
DB CPU Usage	Around 35%	Around 35%

Session - Invalidate

Test to invalidate a user's existing session by calling the session/invalidate API.

Hardware

AM6	
Processor	Intel Xeon® Platinum 8175M @3.1GHz
Cores/vCPU	4
Memory	16 GB
DB - Aurora MySQL 8.0	
Processor	Arm-based AWS Graviton2
Cores/vCPU	4
Memory	32 GB

Configuration

Number of valid sessionToken	50,000

Test result

	w/o Session Persistence	w/ Session Persistence
Num of sessionTokens in CSV	50,000	50,000
Throughput	2496.6 requests/sec	973.3 requests/sec
Avg Resp Time	4 ms	26 ms
AM CPU Usage	Around 90%	Around 90%
DB CPU Usage	Around 15%	Around 60%

SAML - Web Browser SSO - AM6 as IdP

For SAML, CPU utilization is mainly high on AM6 and SP SDK as it involves a lot of XML parsing and RSA operations especially when the operation involves the use of a private key.

In this test, SP SDK is used to simulate the SP side, but the main benchmark is targeted on AM6 server performance.

The test on "SAML only" does not include the authentication. It assumes the test user has a valid session token that can be used to generate the SAML authentication response.

Hardware

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	4 GB
DB - Oracle 12c Standard 2	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	6 GB
SP SDK (as part of jmeter)	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	4 GB

Configuration

SAML signature algorithm	RSA_SHA256
SAML encryption algorithm	AES-256
IdP/SP signing/encryption key	RSA 2048-bit

SAML signing scope	response
Num of users in DB	1000K

```
spec1.setAlgorithm(TokenSpecTO.OATH_ALGO_TOTP);
spec1.setResponseFormatLength(6);
spec1.setTimeInterval(5);//secs
spec1.setSuite("HmacSHA256");
```

Password hash algo	SHA-256
Number of realm accounts	2
Number of login accounts	2
Number of token seeds/blobs per token	2

Test Result

	w/ audit		w/o audit	
SAML Response Encryption	Yes	No	Yes	No
SAML only - Throughput	160 SSO/sec	210 SSO/sec	170 SSO/sec	220 SSO/sec
Include 1FA - Throughput	110 SSO/sec	125 SSO/sec	(not tested)	
Include 2FA - Throughput	70 SSO/sec	70 SSO/sec	(not tested)	

SAML is CPU intensive on AM server and SP SDK.

SAML - Web Browser SSO - SP SDK as SP

For SAML, CPU utilization is mainly high on AM6 and SP SDK as it involves a lot of XML parsing and RSA operations especially when the operation involves the use of a private key.

In this test, the AM6 server is used to simulate the IdP side, but the main benchmark is targeted on SP SDK performance. SP SDK is targeted to use 75% or less of CPU utilization. In this test, SP SDK runs on 2-core server while AM6 is on 8-core. This is to ensure that we can test the SP SDK without the bottleneck of IdP. Thus, this test does not indicate the performance of AM6 server.

Storing of request id and validation of replay attack is not included in this test as it is not in the scope of SP SDK.

Hardware

SP SDK (as part of jmeter)	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	2
Memory	4 GB
AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	8
Memory	4 GB
DB - Oracle 12c Standard 2	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	6 GB

Configuration

SAML signature algorithm	RSA_SHA256
SAML encryption algorithm	AES-256
IdP/SP signing/encryption key	RSA 2048-bit
SAML signing scope	response

Test Result

	Throughput
Signed response	400 SSO/sec
Signed response and encrypted assertion	115 SSO/sec
Signed request, signed response, and encrypted assertion	70 SSO/sec

SAML is CPU intensive on SP SDK

OIDC - Implicit - OAuthProxy and AM6 as OpenID Provider

For OAuth OIDC, CPU utilization is mainly high on AM6, AM Server OIDC operation includes generating access_token and id_token. ID token signing is CPU-intensive due to RSA signing.

In this test, JMeter is used as an OAuth Client or OIDC Relying Party which sends OIDC requests to OAuthProxy as OIDC OpenID Provider endpoints. OAuthProxy acts as a proxy to the AM server where the actual processing happens. This benchmark is targeted for AM server.

Hardware

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	4 GB
DB - Oracle 12c Standard 2	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	6 GB

Configuration

OAuth Client	Single-Page Application (Implicit flow)
No of additional user attribute claims	3
Num of users in DB	3000K
ID Token signing key	RSA-2048

```
spec1.setAlgorithm(TokenSpecTO.OATH_ALGO_TOTP);
spec1.setResponseFormatLength(6);
spec1.setTimeInterval(5);//secs
spec1.setSuite("HmacSHA256");
```

Password hash algo	SHA-256
Number of realm accounts	2

Number of login accounts	2
Number of token seeds/blobs per token	2

Test Result

	w/ audit
OIDC only - Throughput	115 SSO/sec
Include 1FA - Throughput	83 SSO/sec
Include 2FA - Throughput	47 SSO/sec

OAuth OIDC is CPU intensive on AM server due to ID token signing.

Note: OIDC has no audit event. The audit events are for authentication (1FA, 2FA).

OIDC - Authorization Code - OAuthProxy and AM6 as OpenID Provider

For OAuth OIDC, CPU utilization is mainly high on AM6, AM Server OIDC operation includes generating authorization_code, access_token, id_token, and refresh_token. ID token signing is CPU-intensive due to RSA signing.

In this test, JMeter is used as an OAuth Client or OIDC Relying Party which sends OIDC requests to OAuthProxy as OIDC OpenID Provider endpoints. OAuthProxy acts as a proxy to the AM server where the actual processing happens. This benchmark is targeted for AM server.

Hardware

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	4 GB
DB - Oracle 12c Standard 2	
Processor	Intel Xeon E5-2620 v3 @2.4GHz

AM6	
Cores/vCPU	4
Memory	6 GB

Configuration

OAuth Client	Server Side App (Authorization Code flow)
No of additional user attribute claims	3
Num of users in DB	3000K
ID Token signing key	RSA-2048

spec1.setAlgorithm(TokenSpecTO.OATH_ALGO_TOTP);
spec1.setResponseFormatLength(6);
spec1.setTimeInterval(5);//secs
spec1.setSuite("HmacSHA256");

Password hash algo	SHA-256
Number of realm accounts	2
Number of login accounts	2
Number of token seeds/blobs per token	2

Test Result

	w/ audit
OIDC only - Throughput	100 SSO/sec
Include 1FA - Throughput	79 SSO/sec
Include 2FA - Throughput	47 SSO/sec

OAuth OIDC is CPU intensive on AM server due to ID token signing.

> **Note:** OIDC has no audit event. The audit events are for authentication (1FA, 2FA).

OIDC - Refresh Token - OAuthProxy and AM6 as OpenID Provider

For OIDC Refresh Token, CPU utilization is mainly high on AM server, it involves generating access_token, id_token and refresh_token. ID token signing is CPU-intensive due to RSA signing.

In this test, JMeter is used as an OAuth Client or OIDC Relying Party which sends OIDC requests to OAuthProxy as OIDC OpenID Provider endpoints. OAuthProxy acts as a proxy to the AM server where the actual processing happens. This benchmark is targeted for AM server.

Hardware

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	4 GB
DB - Oracle 12c Standard 2	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	6 GB

Configuration

OAuth Client	Server Side App (Authorization Code flow)
No of additional user attribute claims	3
ID Token signing key	RSA-2048

Test Result

OIDC refresh_token only - Throughput	153 requests/sec

OAuth OIDC is CPU intensive on AM server due to ID token signing.

OIDC - Relying Party

In this test, AM API is used to call OIDC RP and OIDC OP. AM as OIDC OP is used in this testing. However, this benchmark is targeted to stress test the AM as OIDC RP, not the AM as OIDC OP.

Hardware

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	4 GB
DB - Oracle 12c Standard 2	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	6 GB

Configuration

Grant Type	Authorization Code flow
Token Endpoint Authentication Method	ClientSecret
User Mapping Lookup Option	User Id
ID Token is JWE Encrypted	true
Encryption Key Algorithm	EC-P521
Num of users in DB	100K
AM version	5.7.2.7216-GA

Test Result

Request	Throughput
preauthenticate to generate nonce	85 requests/sec

Request	Throughput
process authorization code	85 requests/sec

Notes: The bottleneck is at the AM CPU because the test is CPU-intensive.

OIDC - Introspection Endpoint - OAuthProxy and AM6 as OpenID Provider

In this test, JMeter is used as an OAuth Client which sends OIDC requests (introspect) to OAuthProxy as OIDC OpenID Provider endpoints. OAuthProxy acts as a proxy to the AM server where the actual processing happens.

Hardware

OAuthProxy	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	4 GB
AM6 - OIDC OP	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	4 GB
DB - Oracle 12c Standard 2	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	6 GB

Configuration

OAuth Client	Server Side App (Authorization Code flow)
Access Token type	Opaque/JWT

Test Result

Request	Throughput
introspect - JWT	475 requests/sec
introspect - Opaque	473 requests/sec

Native WSA Reverse Proxy - URL Authorization

This test includes WSA URL authorization check when an HTTP request to an application arrives at WSA Reverse Proxy. WSA will evaluate whether to allow/disallow the request based on the authorization policies and the user's session.

Hardware

WSA	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	2
Memory	4 GB
AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	4 GB
DB - Oracle 12c Standard 2	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	6 GB

Configuration

Session persistence	on
No of applications	3
No of URL patterns	5

No of attributes returned in headers	5

Test Result

When WSA receives a request for the first time for that particular user session, WSA calls AM server to do the authorization check (AM will also audit this request). At the same time, WSA caches the access policy. For subsequent requests, WSA uses the access policy cached in WSA to evaluate the requests locally to increase performance. On certain interval (WSA refresh window), WSA calls AM server to refresh the user's session.

The result below is categorized to the two phases: when AM is the one processing the authorization and when WSA is the one processing the authorization.

Phase	Throughput	WSA CPU usage	AM CPU usage	DB CPU usage
Authorization check by AM	350 reqs/sec	40%	80%	90%
Authorization check by WSA	820 reqs/sec	90%	7%	7%

From the result, WSA can process efficiently with just two vCPU using its cached policies. This reduces the load in AM server significantly.

2.4. USO

USO Login & App Info Download

The testing is to find out the performance figure for USO on AM server. Test scenario includes simulation of user login via CLP (using SystemPasswordBasic), querying user's USO entitlements and downloading the USO application information.

Hardware

AM6	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	4 GB
DB - Oracle 12c Standard 2	
Processor	Intel Xeon E5-2620 v3 @2.4GHz
Cores/vCPU	4
Memory	6 GB

Configuration

No of USO applications	60
No of USO users	10000
No of USO application entitled to each user	60

All other settings are left to default, particularly the "Download Mode" is "Download per login".

Test Result

Test includes this sequence:

- Login via CLP (system password basic) to USO
- Download USO properties
- Refresh session

- Download USO application information for all apps entitled to the user

EnablePSE	Throughput
No	5.8 per sec
NoPwd	6.5 per sec
Yes	6.5 per sec

AM server's CPU usage is around 65% while DB's CPU usage is around 35%.